# DIBYA JOY PAUL

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GitHub | AI Projects | AI Notes | LinkedIn

## PROFILE

AI Engineering master's student and Mechatronics graduate building agentic AI, RAG/NLP, computer-vision, and data-driven systems. Recent work includes an energy-aware prompt-optimization benchmark with MLflow and statistical evaluation, plus a public AI-engineering knowledge base covering LangGraph, tool use, distributed systems, cloud deployment, SQL, transfer learning, and LLM cost/latency.

## EDUCATION

**UNITED INTERNATIONAL UNIVERSITY** Dhaka, Bangladesh
M.Sc. in Computer Science & Engineering (Major: AI Engineering) May 2025 - Present
**RAJSHAHI UNIVERSITY OF ENGINEERING & TECHNOLOGY (RUET)** Rajshahi, Bangladesh
B.Sc. in Mechatronics Engineering Jan 2018 - Oct 2023

## TECHNICAL SKILLS

**Programming & Data:** Python, SQL/PostgreSQL/SQLite, C/C++, JavaScript, HTML/CSS; Pandas, NumPy, Matplotlib, Seaborn, SciPy
**AI/ML:** Machine Learning, Deep Learning, NLP, LLMs, Agentic AI, LangGraph, RAG, embeddings, Transfer Learning, Computer Vision, SVM, XGBoost
**AI Engineering & MLOps:** FastAPI, Pydantic, Streamlit, OpenAI API, tiktoken, ChromaDB, function/tool calling, SQLAlchemy, MLflow, Pytest, GitHub Actions
**Systems & Cloud:** Docker, REST/async streaming, distributed-systems fundamentals, AWS/GCP fundamentals, system design, observability, Git/GitHub

## SELECTED AI ENGINEERING PROJECTS

**ECOPROMPT - ENERGY-AWARE PROMPT OPTIMIZATION** GitHub
- Built an agentic benchmark comparing baseline, rule-based, LLM-rewrite, and self-correcting tool-use strategies across 11 prompts and four downstream models (176 experiment cells), with exact token counting and modelled cost, energy, and CO2e metrics.
- Implemented custom count_tokens, rewrite_prompt, and estimate_energy tools; LLM-as-judge quality scoring; MLflow/CSV tracking; Pytest; and Wilcoxon, ANOVA, Kruskal-Wallis, and Pareto analysis. Offline results retained 0.92-0.99 quality with 13.9-28.7% token reduction.

**BENVENUTO - AGENTIC AI GUIDE FOR INTERNATIONAL STUDENTS** GitHub
- Built a FastAPI and React/Vite assistant with OpenAI function calling, server-sent-event streaming, ChromaDB RAG, RSS, OpenStreetMap/Overpass, web search, vision, Whisper, and text-to-speech.

**BANGLA ORDER AGENT** GitHub
- Developed a Bangla/Banglish/English NLP order pipeline with validation, inventory and pricing checks, SQLite persistence, HTML invoices, FastAPI/Streamlit interfaces, Docker, Pytest, and GitHub Actions CI.

**SEMANTIC SEARCH VS. KEYWORD SEARCH** GitHub
- Created a FastAPI movie-search application comparing keyword retrieval with OpenAI embeddings indexed in ChromaDB, including nearest-neighbor similarity scoring and side-by-side browser results.

**TOOL-CALLING AGENT** GitHub
- Implemented a multi-step tool-selection loop for calculator, weather, and time functions, with restricted-AST math parsing, visible tool traces, and a six-round guard against unsafe or runaway execution.

**AI ENGINEERING NOTES - TECHNICAL KNOWLEDGE BASE** GitHub
- Authored 10 implementation-oriented Jupyter notebooks on OpenAI APIs and async streaming, LangGraph workflows, tool schemas and execution loops, PostgreSQL, distributed systems, cloud deployment, Git, transfer learning, and LLM token cost/latency.
- Documented production patterns including state and persistence, human-in-the-loop approvals, retries/timeouts/idempotency, latency budgets, observability, AWS/GCP serving options, and feature extraction, fine-tuning, LoRA, and PEFT.

## RESEARCH EXPERIENCE

**BEYOND SOFTMAX - CATTLE DISEASE DETECTION BENCHMARK** 2025
- Benchmarked transfer-learned CNN and Vision Transformer representations with neural, SVM, and XGBoost classifiers for image-based cattle disease detection.

**CLIMATE-RELATED RESPIRATORY DISEASE RISK ANALYSIS** 2025
- Applied logistic regression to discover climate-related risk thresholds and examine cross-income transferability of respiratory-disease risk models.

**UNDERGRADUATE THESIS - DIGITAL TWIN PREDICTIVE MAINTENANCE** Oct 2022 - Jun 2023
- Developed an approach for a digital-twin-assisted fault detection and isolation system for predictive maintenance of pneumatic systems.

**INDUSTRY 5.0 HUMAN-ROBOT INTERACTION CASE STUDY** Jan 2022 - Oct 2022
- Reviewed enabling technologies, technical challenges, and future prospects for next-generation human-robot interaction in Industry 5.0.

## WORK EXPERIENCE

**GRADUATE RESOURCE ENHANCING CENTRE (GREC)** Dhaka, Bangladesh
Tutor - GRE Quantitative and Verbal Reasoning Jun 2024 - Jul 2025
- Delivered five classes per week and prepared structured lecture materials and practice sheets for quantitative and verbal reasoning.
- Conducted two mock tests per month, evaluated performance, and provided targeted feedback to strengthen analytical reasoning and test strategy.

## ADDITIONAL

**Earlier Engineering Projects:** Nvidia Jetson Nano self-driving car; NodeMCU IoT attendance counter; IR/ultrasonic line-following and wall-guided robot
**Standardized Tests: IELTS** 8.0 (L 9, R 8, W 7, S 7); **GRE**: 320 (Quant 162, Verbal 158, AWA 3)
**Research Interests:** Agentic AI, AI chatbots, RAG/NLP, intelligent automation, scalable AI systems, MLOps, predictive maintenance, and AI for robotics
**Portfolio:** Source code and technical notes are linked in the header and project titles.